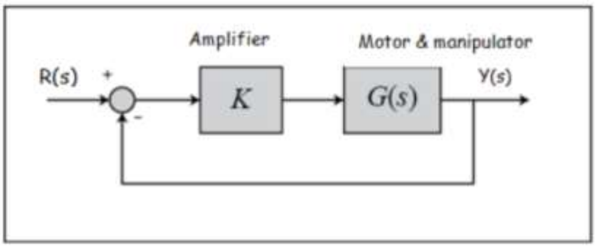
ME 15- 1403 MECHATRONICS

Max Marks: 50 Time: 135 min

(Problem statement for Q 1-5)

The use of laser for surgery requires high accuracy for position and velocity response (or step and ramp inputs). The figure shows the block diagram of a laser surgery system, which uses a dc motor and manipulator for the laser.

The transfer function $G(s) = \frac{50}{s(s+10)(s+5)}$



Q1. Determine the range of values of the amplifier gain K for which the system will be stable. (15 min) (5 marks)

Q2. What should be the amplifier gain K so that the steady state error for a ramp input $r(t) = At$ (where $A = 1\ mm/s$) should be less than or equal to $0.1\ mm$ while remaining in stable condition? (15 min) (5 marks)

Q3. Obtain the state spce representation of the system using the value of K in Q2. (15 min) (5 marks)

Q4. Plot the root locus of the system clearly describing each step. Find the closed loop poles of the system by considering the gain K obtained in Q2. Is it possible to justify the dominance of the second order complex conjugate poles?If so, calculate the percentage overshoot and 2% settling time of the system. Design a suitable controller if the settling time has to be less than $2s$. (45 min) (15 marks)

Q5. Draw the Bode diagram of the system using the amplifier gain obtained in Q2. Identify the gain margin and phase margin. How much change can be possible in gain K before the system become unstable? Compare your results with Q1. (25 min) (10 marks)

Q.6 An automatic door senses when a person comes near using an ultrasound sensor. The door opens and after a delay of 10s it closes. The door is opened and closed using a double acting hydraulic cylinder which is controlled using a two solenoid 5/3 DCV. Draw the ladder diagram.

(20 min) (10 mark)

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<td>View from top

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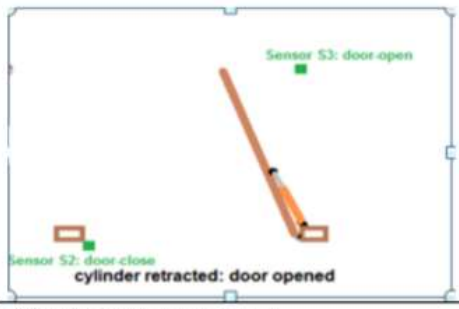
<td>
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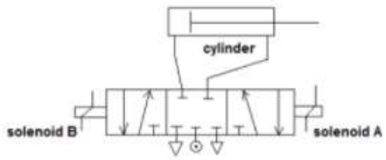
<td>
</td>
<td>X0- ultrasound sensor S1
x2- door close sensor S2
X3- door open sensor S3
Y0- solenoid a is connected to output port Y0
Y1- solenoid b is connected to output port Y1</td>
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</table>